Handwriting synthesis: deployable AI model development

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In our project, we treat handwriting synthesis as a sequence prediction problem for the AI model. For this task, we initially chose an RNN model, that was created following Alex Graves' 2013 Paper "**Generating Sequences With Recurrent Neural Networks" [1].** Although our initial model was successful in generating unique handwriting styles, completing one of our requirements, the quality of the handwriting were not satisfying from much of the generated samples. The model presented a few issues in the generated output. We attempted to improve the model by solving two of the main issues.

Architecture of the model

The model we used is mainly a Recurrent Neural Network (RNN). This type of network was chosen because there ability to capture sequential information well. The model architecture in the highest level looked like this:

input → GMM attention window → LSTM gate (RNN) → MDN network → GMM sampled output

The GMM (Gaussian Mixture Model) based attention mechanism uses a window over the sentences being read, and learns to predict the probability of each character it reads to be a certain letter. The LSTM cells (RNN cells) learn to predict parameters that define the Gaussians of the probabilities.

The problem

The main two issues we identified from our generated outputs:

1. Style inconsistency: The handwriting synthesis model we use generated style sampled output by priming the output. The priming process is done by first concatenating the input strings with the sample style, then forcing the model to sample from the style to be copied. As the model samples, the style is encoded into the LSTM's hidden state.

   This priming process has two shortcomings, noted by Graves-

   - Style decay - The LSTM's hidden state eventually changes as the model continues to sample.
   - No explicit style control - The LSTM learns implicitly from the style, giving the user no control.

   The result was that some of the outputs had the same characters written in different ways. Some of the outputs had a completely different styling by the time the sentence ended.

2. Skipping/stalling characters: Some of the output sequences skipped some of the characters. For some of the outputs, some characters were repeated multiple times.

   This was likely due to the attention mechanism window processing more than one characters at a time.

Solution

To solve these issues, we first attempted to perform a Grid search of hyperparameters for the model. For training this model, we got the help of Cristian Dutescu (Our sponsor from Orion Defense Solutions), who modified the model training code to make it trainable in parallel on 4 GPUs.

The hyperparameter grid search did not yield any improvements on these issues. The generated outputs had their own issues (grainy text).

Next, we found some improvements to the components of the neural network architecture. The first technique is the Variational Auto Encoder technique used on the paper “**DeepWriting: Making Digital Ink Editable via Deep Generative Modeling”[2].** The model builds on the Graves model, specifically how the styling is incorporated. Instead of priming through the LSTM’s hidden state, the paper introduces a latent style embedding. A new bi-LSTM layer was used to create this style embedding.

Another method was to include the GMMv2b, a modification of the GMM attention mechanism by Battenbarg et. al. [3]. Three modifications of the mechanism:

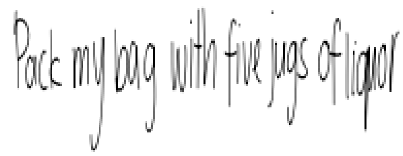
- Using the exp() equation for the three parameters
- Bias initialized from a non-zero value
- Softplus equation for alpha parameters

The final method was the stepwise monotonic attention [4], which made the attention window either stay at the current character or move to the next at each timestep.

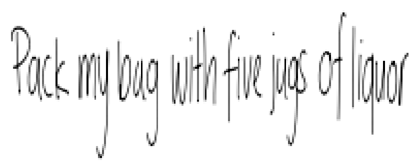
Results

The VAE and GMMV2b techniques were successful in takling the issues mentioned above. The variational autoencoder was great at maintaining consistency in the styles used.

Variational Auto Encoder outputs:

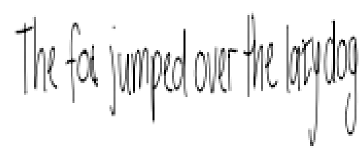


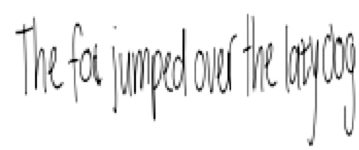
Text box input:



Text box input:


The fox jumped over the lazy dog


The fox jumped over the lazy dog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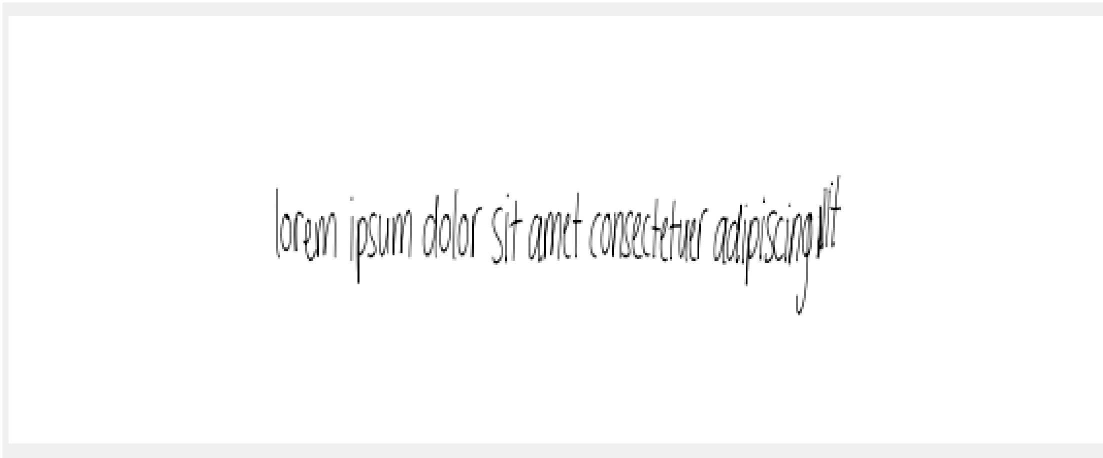
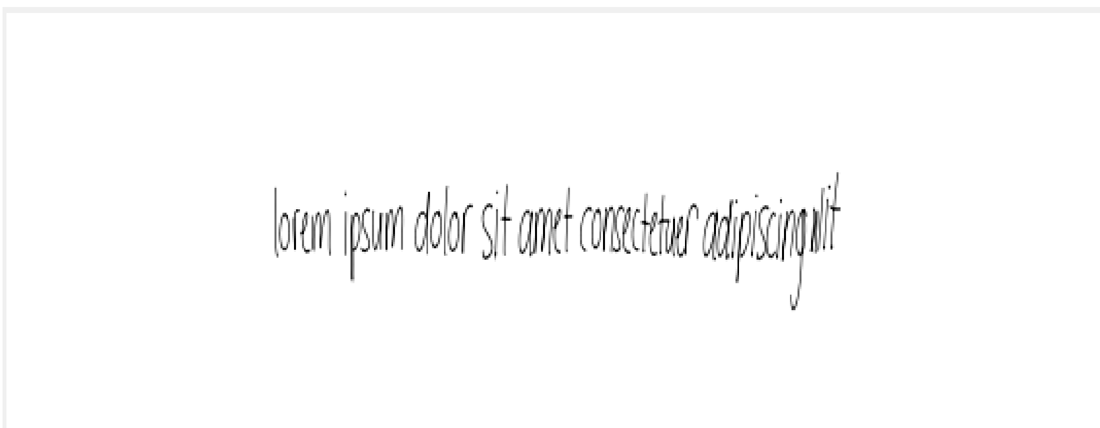
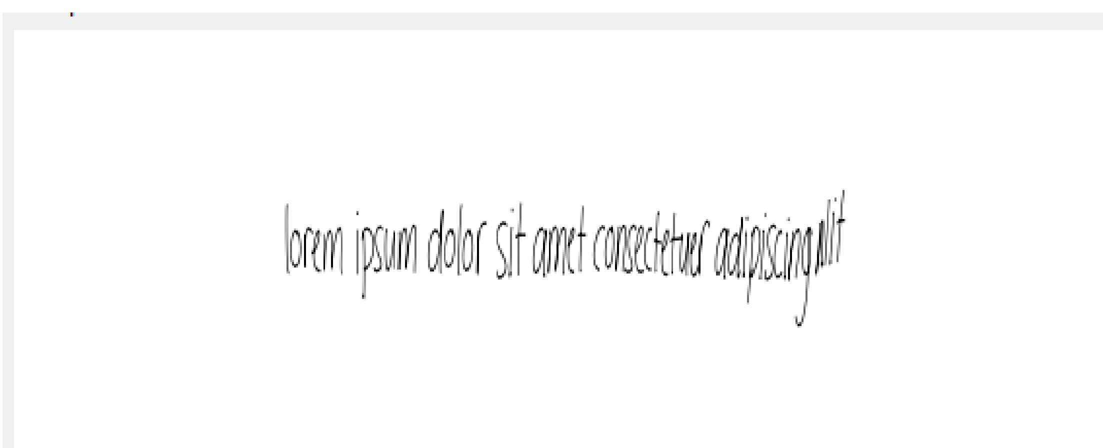
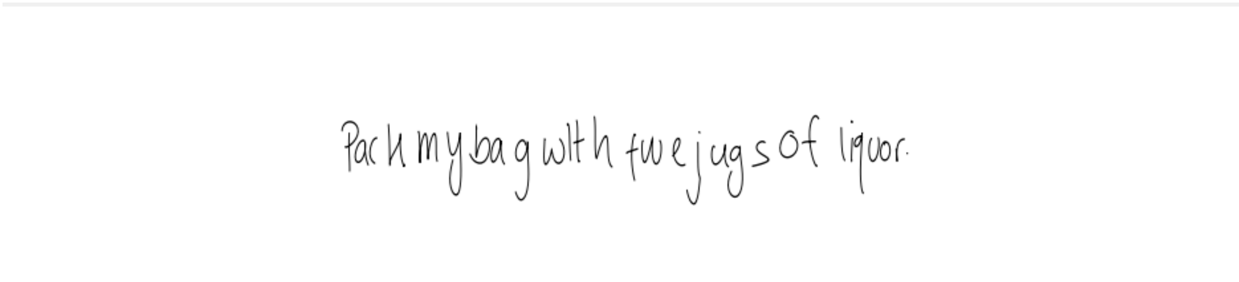


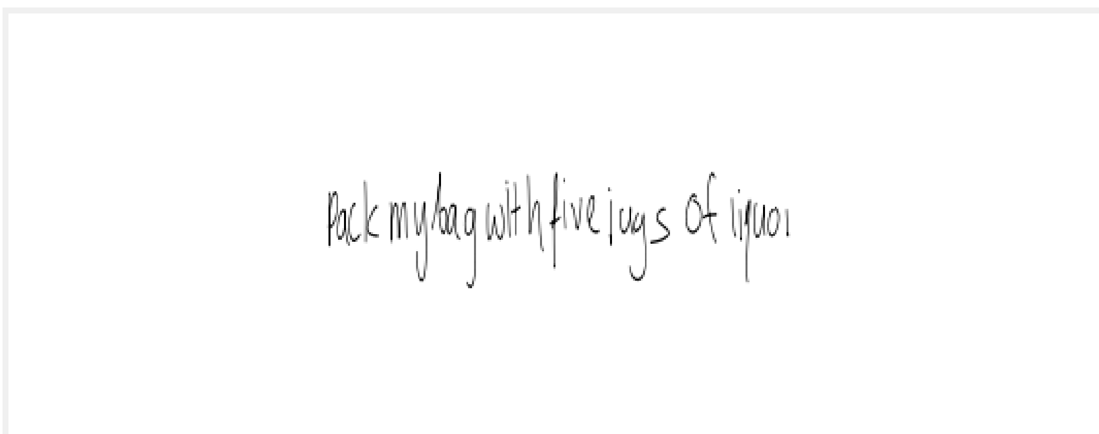


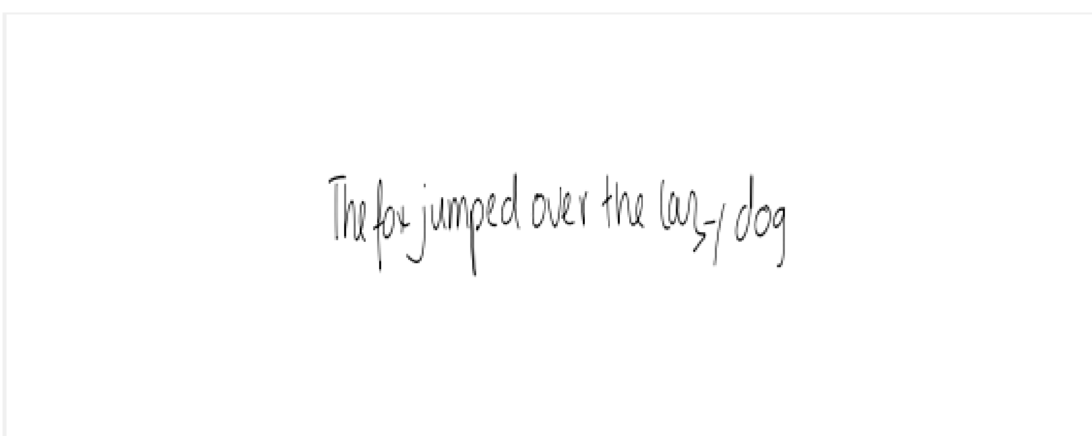


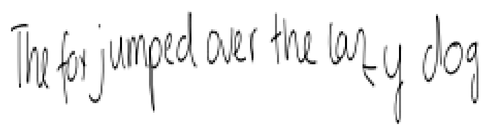
These samples were all generated from the same style with different bias values.

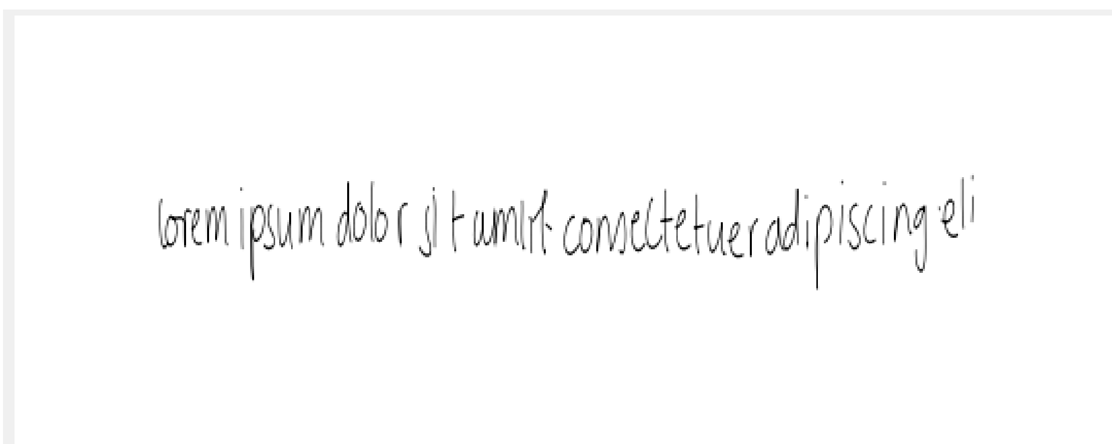
The GMMV2b was able to prevent skips and stalls. The following examples are from the GMMV2b model:











Original outputs from the GMM included model (with skipped and repeating characters):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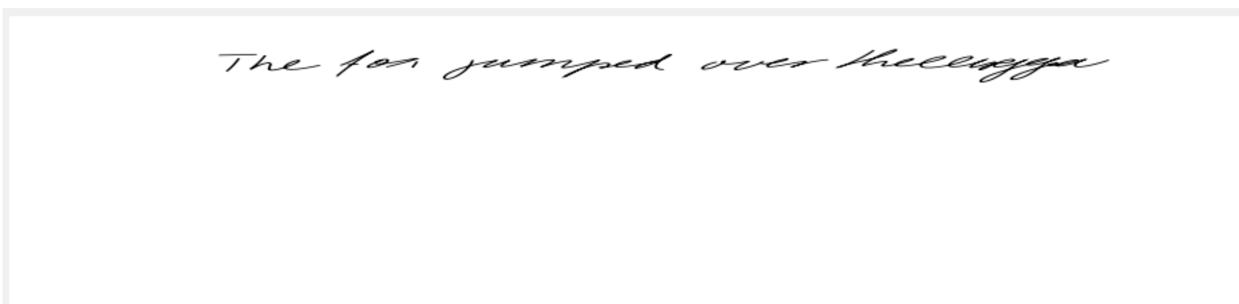
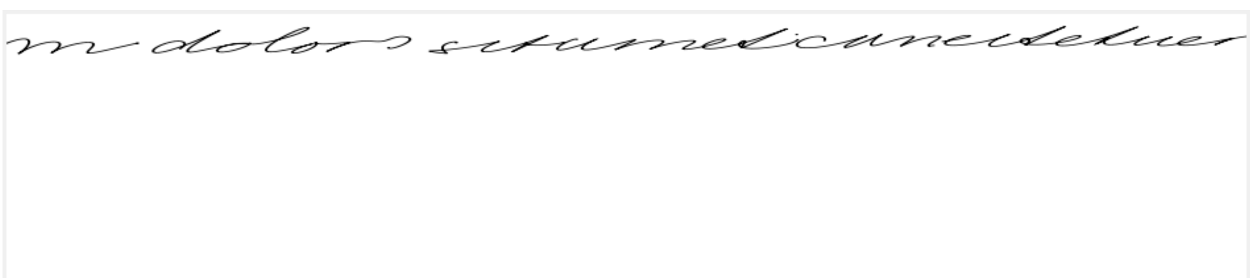
jumped over the


The fox jumped over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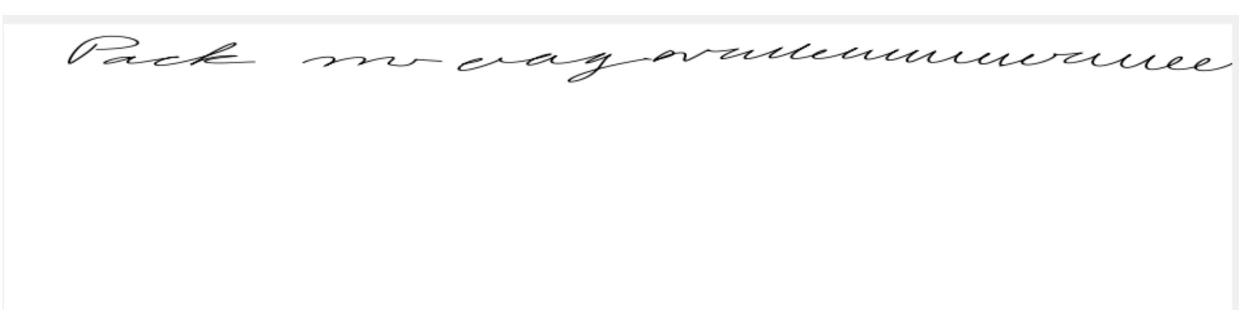
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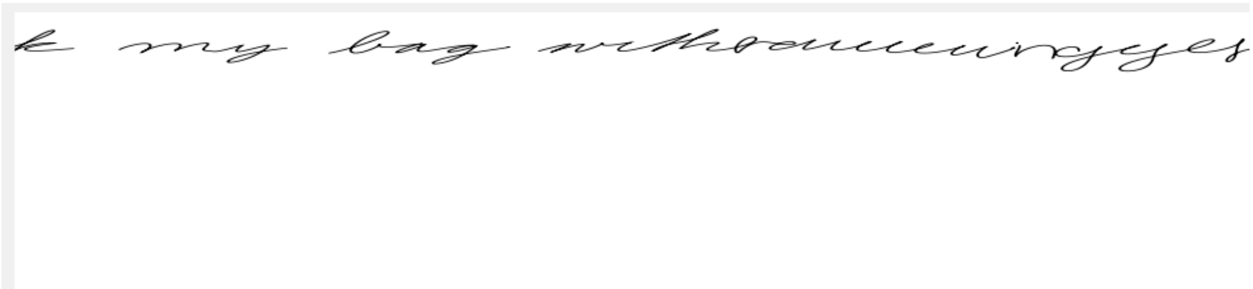

The SMA model did not produce satisfactory results as the outputs generated had swapped characters or entirely new characters that were not part of the original input. SMA was originally developed for TTS (Text to speech) tasks, although it was inspired by the Graves model.

Conclusion:

Through iterative research, implementation and testing, we were able to improve the original graves model for deployment in our handwriting generation system. Through these improvements, the model was more successful in completing the requirements for our project.

In the future, we hope to combine the VAE and GMMV2b techniques to be used in the production model of our handwriting generation system. The model is also made trainable for future improvements.

References:

1. A. Graves, "Generating sequences with recurrent neural networks," *arXiv preprint arXiv:1308.0850*, 2013.
2. Aksan, F. Pece, and O. Hilliges, "DeepWriting: Making digital ink editable via deep generative modeling," in *Proc. CHI*, 2018, pp. 1–14.

3. E. Battenberg, R. J. Skerry-Ryan, S. Mariooryad, D. Stanton, D. Kao, M. Shannon, and T. Bagby, "Location-relative attention mechanisms for robust long-form speech synthesis," in *Proc. ICASSP*, 2020, pp. 6194–6198.
4. M. He, Y. Deng, and L. He, "Robust sequence-to-sequence acoustic modeling with stepwise monotonic attention for neural TTS," in *Proc. Interspeech*, 2019, pp. 1293–1297.